Heat is not temperature

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish temperature from energy transferred because of a temperature difference.

SUCCESS CRITERIA - I can...

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “heat” and “temperature” to explain your answer. Sentence starters are optional.

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

thermal energy: internal energy associated with microscopic motion/interactions

01 Warm-up: bring back an earlier idea

A simplified calculation requires **15 J** for escape. An object has only **12 J** and receives no energy later. Does the calculation predict escape? Explain.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q = mc\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

Worked reasoning: A cup and a bathtub can both be at **40°C**. Equal temperature does not imply equal internal energy: amount and type of material also matter.

Your turn: 1. **Draw equal temperature indicators but** different amounts of water.
2. **Predict heat-flow direction** if the water samples are brought into thermal contact at the same temperature.

Sentence starter: Both ... . They differ in ... because ... .

Sample A



Sample B



Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cup and a bathtub contain the same substance at **40°C**. Are their temperatures equal? Must their total internal energies be equal?

Sentence starter: I notice ... . This supports ... because ... .



Cup: 40°C



Tub: 40°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

A metal spoon at **60°C** is placed in water at **20°C**.
1. **Label both samples** and **draw the initial energy-transfer arrow**.
2. **Predict which temperature rises** and which falls.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Spoon: 60°C



Water: 20°C

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wheel rolls down a ramp, losing **24 J** gravitational energy. Its center-of-mass kinetic energy gain is **20 J** with negligible losses. What energy term is missing? **Show the energy changes** and transfers; **label units**.

Word help: negligible = small enough to leave out of this model

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial | **Transfers** | **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

1. **Label a thermometer reading** and an energy-transfer arrow on the diagram.
2. **Explain one difference** between temperature and heat, including the usual units for each.

Sentence starter: Both … . They differ in … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A small metal spoon is hotter than a large bath. Can temperature alone tell which contains more internal energy? **Explain what else you would** need.

Sentence starter: I think … . I can point to …, which shows …, so … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3